Banik Assignment 4

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```
setwd('/Users/arpabanik/Library/CloudStorage/OneDrive-BentleyUniversity/R-Projects/MA 347/Assignment 4')

#Import necessary libraries
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
library(reshape2)
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

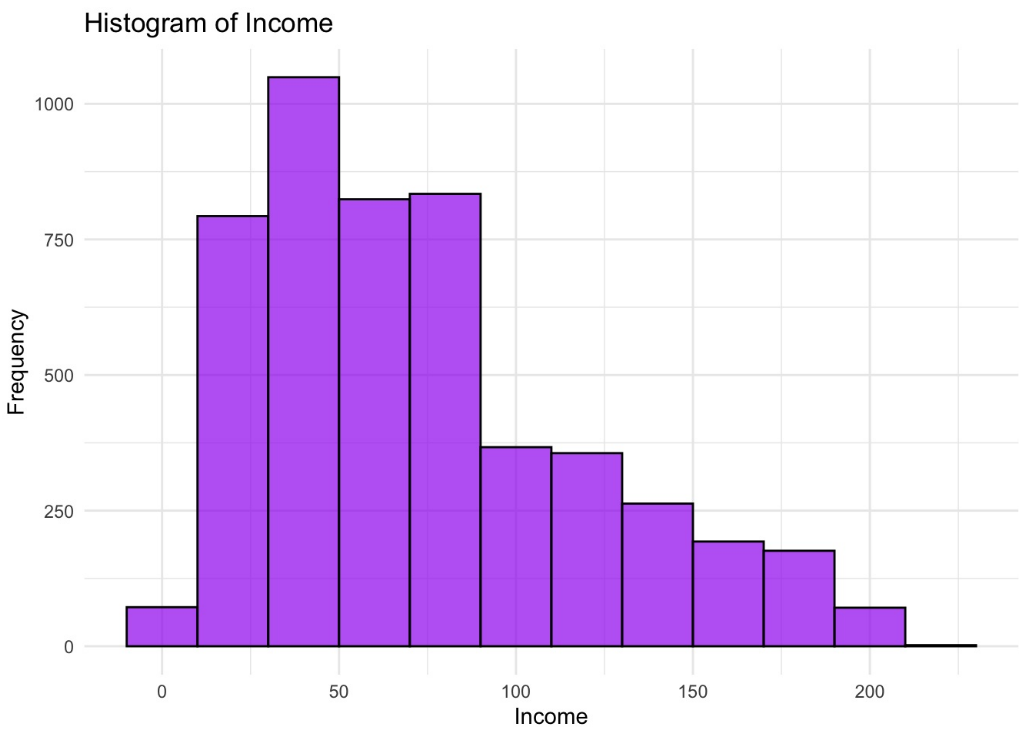
```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(e1071)
data<-read.csv('/Users/arpabanik/Library/CloudStorage/OneDrive-BentleyUniversity/R-Projects/MA 347/Assignment 4/U
niversalBank (2).csv')
```

Problem 1

```
# 1. Create a histogram for a quantitative variable (e.g., 'Income')
ggplot(data, aes(x = Income)) +
  geom_histogram(binwidth = 20, fill ="purple", color = "black", alpha = 0.7, position = "identity") +
  labs(title = "Histogram of Income", x = "Income", y = "Frequency") +
  theme_minimal()
```




```
mean(data$Income)
```

```
## [1] 73.7742
```

The plot reveals a clear trend in the distribution of customer income within the range of 25 to 50, where the frequency is highest, exceeding 1000 customers. As income increases beyond this range, the frequency steadily decreases, indicating a declining proportion of customers. The overall distribution of income is right-skewed, with a mean income of approximately 73.7742. This skewness suggests that a majority of customers have relatively lower incomes, while a smaller proportion exhibits higher income levels. Additionally, the plot underscores the concentration of customers with incomes between 25 and 50, hinting at a potential target demographic for the bank's personal loan campaign within this income bracket.

2. Create a data frame where CreditCard=1 and Online=1

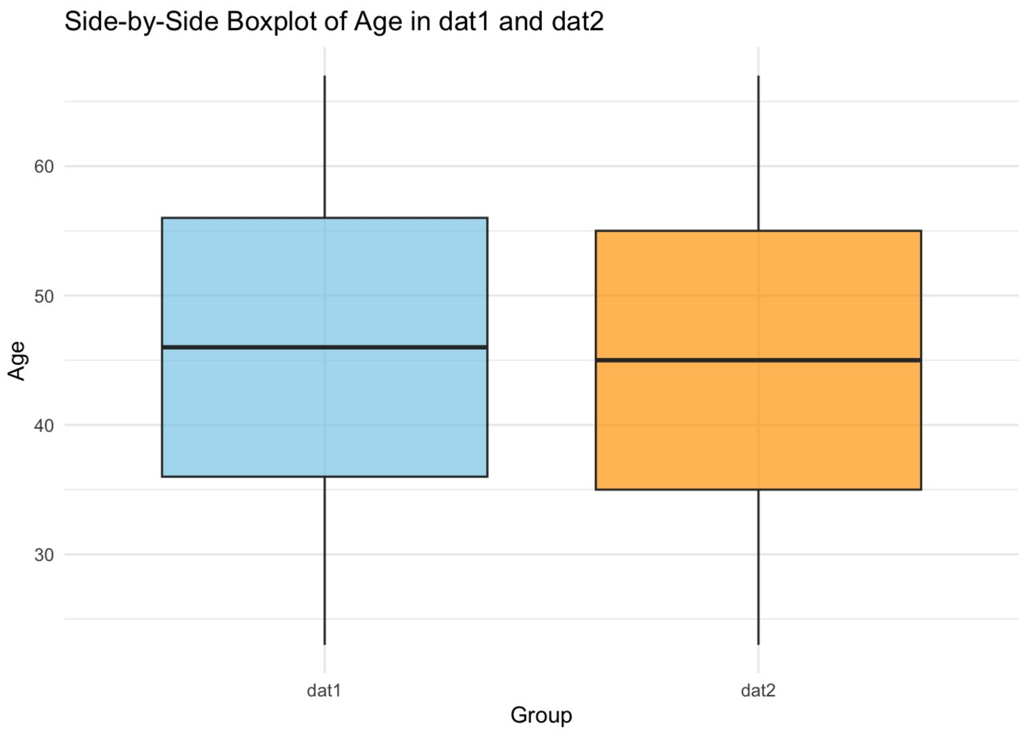
```
dat1 <- filter(data, CreditCard == 1, Online == 1)
```

3. Create a data frame where CreditCard=0 and Online=0

```
dat2 <- filter(data, CreditCard == 0, Online == 0)
```

4. Create a side-by-side boxplot for Age in dat1 and dat2

```
ggplot() +
  geom_boxplot(data = dat1, aes(x = "dat1", y = Age), fill = "skyblue", alpha = 0.7) +
  geom_boxplot(data = dat2, aes(x = "dat2", y = Age), fill = "orange", alpha = 0.7) +
  labs(title = "Side-by-Side Boxplot of Age in dat1 and dat2", x = "Group", y = "Age") +
  theme_minimal()
```




```
summary(dat1)
```

```
##        ID            Age          Experience        Income      
##  Min.   :  16   Min.   :23.00   Min.   :-2.00   Min.   :  8.00  
##  1st Qu.:1275   1st Qu.:36.00   1st Qu.:11.00   1st Qu.: 38.00  
##  Median :2488   Median :46.00   Median :21.00   Median : 66.50  
##  Mean   :2529   Mean   :45.68   Mean   :20.48   Mean   : 74.95  
##  3rd Qu.:3806   3rd Qu.:56.00   3rd Qu.:30.75   3rd Qu.: 97.25  
##  Max.   :5000   Max.   :67.00   Max.   :43.00   Max.   :224.00  
##     ZIP.Code         Family          CCAvg          Education   
##  Min.   : 9307   Min.   :1.000   Min.   :0.000   Min.   :1.00  
##  1st Qu.:92037   1st Qu.:1.000   1st Qu.:0.700   1st Qu.:1.00  
##  Median :93943   Median :2.000   Median :1.500   Median :2.00  
##  Mean   :93195   Mean   :2.431   Mean   :1.918   Mean   :1.82  
##  3rd Qu.:94611   3rd Qu.:3.750   3rd Qu.:2.500   3rd Qu.:3.00  
##  Max.   :96150   Max.   :4.000   Max.   :8.800   Max.   :3.00  
##     Mortgage      Personal.Loan     Securities.Account   CD.Account   
##  Min.   :  0.00   Min.   :0.00000   Min.   :0.00000    Min.   :0.000  
##  1st Qu.:  0.00   1st Qu.:0.00000   1st Qu.:0.00000    1st Qu.:0.000  
##  Median :  0.00   Median :0.00000   Median :0.00000    Median :0.000  
##  Mean   : 56.31   Mean   :0.09297   Mean   :0.09751    Mean   :0.254  
##  3rd Qu.:103.00   3rd Qu.:0.00000   3rd Qu.:0.00000    3rd Qu.:1.000  
##  Max.   :587.00   Max.   :1.00000   Max.   :1.00000    Max.   :1.000  
##      Online    CreditCard
##  Min.   :1   Min.   :1   
##  1st Qu.:1   1st Qu.:1   
##  Median :1   Median :1   
##  Mean   :1   Mean   :1   
##  3rd Qu.:1   3rd Qu.:1   
##  Max.   :1   Max.   :1   
```

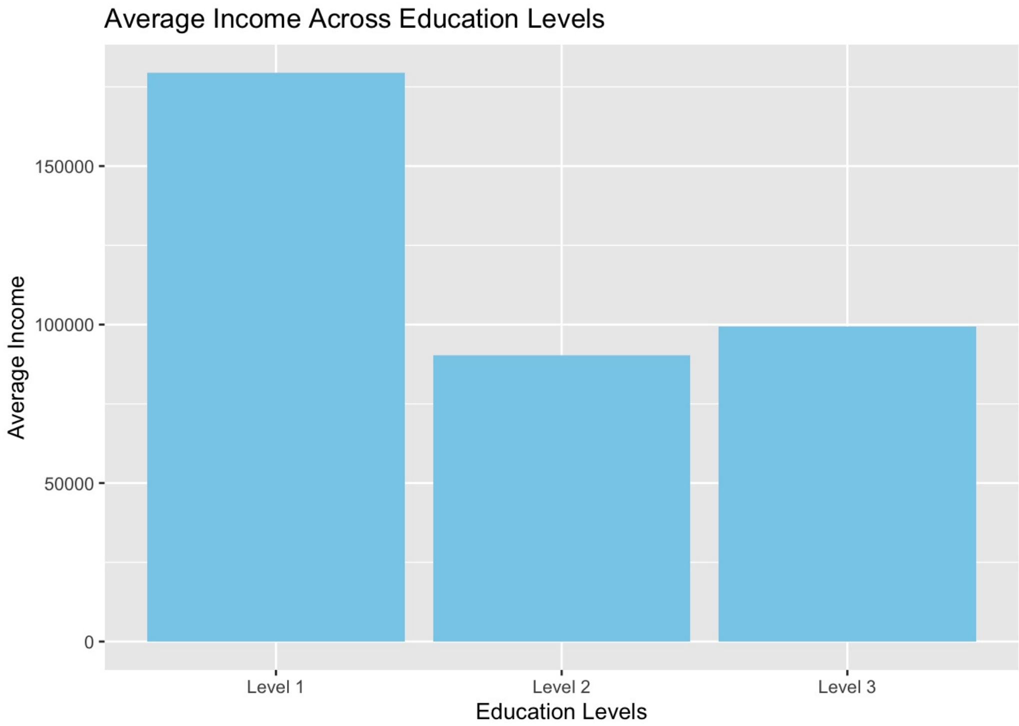
```
summary (dat2)
```

```
##        ID            Age          Experience       Income          ZIP.Code    
##  Min.   :   1   Min.   :23.00   Min.   :-3.0   Min.   :  8.00   Min.   :90005  
##  1st Qu.:1277   1st Qu.:35.00   1st Qu.:10.0   1st Qu.: 38.00   1st Qu.:91774  
##  Median :2483   Median :45.00   Median :20.0   Median : 64.00   Median :93117  
##  Mean   :2485   Mean   :45.14   Mean   :19.9   Mean   : 73.55   Mean   :93091  
##  3rd Qu.:3703   3rd Qu.:55.00   3rd Qu.:30.0   3rd Qu.: 98.25   3rd Qu.:94572  
##  Max.   :4998   Max.   :67.00   Max.   :43.0   Max.   :203.00   Max.   :96651  
##      Family          CCAvg          Education       Mortgage    
##  Min.   :1.000   Min.   : 0.000   Min.   :1.00   Min.   :  0.0  
##  1st Qu.:1.000   1st Qu.: 0.700   1st Qu.:1.00   1st Qu.:  0.0  
##  Median :2.000   Median : 1.600   Median :2.00   Median :  0.0  
##  Mean   :2.376   Mean   : 1.955   Mean   :1.88   Mean   : 58.6  
##  3rd Qu.:3.000   3rd Qu.: 2.500   3rd Qu.:3.00   3rd Qu.:102.0  
##  Max.   :4.000   Max.   :10.000   Max.   :3.00   Max.   :617.0  
##  Personal.Loan     Securities.Account   CD.Account           Online 
##  Min.   :0.00000   Min.   :0.0000     Min.   :0.000000   Min.   :0  
##  1st Qu.:0.00000   1st Qu.:0.0000     1st Qu.:0.000000   1st Qu.:0  
##  Median :0.00000   Median :0.0000     Median :0.000000   Median :0  
##  Mean   :0.08964   Mean   :0.1008     Mean   :0.002101   Mean   :0  
##  3rd Qu.:0.00000   3rd Qu.:0.0000     3rd Qu.:0.000000   3rd Qu.:0  
##  Max.   :1.00000   Max.   :1.0000     Max.   :1.000000   Max.   :0  
##    CreditCard
##  Min.   :0   
##  1st Qu.:0   
##  Median :0   
##  Mean   :0   
##  3rd Qu.:0   
##  Max.   :0   
```

The boxplots reveal that the age distributions in “dat1” and “dat2” are quite comparable. The median age in “dat1” is slightly higher at 46.00, compared to 45.00 in “dat2.” Both datasets exhibit similar means (45.68 in “dat1” and 45.14 in “dat2”) and show symmetric distributions, as indicated by the closeness of mean and median values. The minimum and maximum ages are identical for both datasets. The boxplots also show overlapping interquartile ranges (IQRs) and median lines, suggesting a high degree of similarity in the central tendencies of the age distributions. Any potential differences in the medians might be subtle, with “dat1” having a slightly higher median age, confirming the numeric observations.

5. Create a bar plot to compare average Income across Education levels

```
# Create a bar plot
ggplot(data, aes(x = factor(Education), y = Income)) +
  geom_bar(stat = "identity", fill = "skyblue") +
  labs(title = "Average Income Across Education Levels",
       x = "Education Levels",
       y = "Average Income") +
  scale_x_discrete(labels = c("1" = "Level 1", "2" = "Level 2", "3" = "Level 3"))
```



In the bar plot comparing average income across three levels of education, it is evident that individuals with education level 1 exhibit the highest average income of about 85000, marked by a bar significantly above the others. On the other hand, education levels 2 and 3 are comparable, with level 3 (66000 average income) slightly edging out level 2 (64000 average income) in terms of average income. This bar plot underscores the positive correlation between higher education levels and average income, highlighting the financial advantage associated with a more advanced educational background.

```
Problem 2
```

1. Partition the data into training (60%) and validation (40%) sets

```
set.seed(123)
train_indices <- sample(1:nrow(data), 0.6 * nrow(data))
train.df <- data[train_indices, ]
validation.df <- data[-train_indices, ]
```

2. Create a pivot table for the training data

```
ftable(train.df$CreditCard, train.df$Personal.Loan, train.df$Online)
```

```
##          0    1
##
## 0 0    785 1145
##   1     65  122
## 1 0    317  475
##   1     34   57
```

3. Probability of loan acceptance given CC=1 and Online=1

```
prob_loan_given_CC1_Online1 <- sum(train.df$Personal.Loan == 1 & train.df$CreditCard == 1 & train.df$Online == 1)
/ sum(train.df$CreditCard == 1 & train.df$Online == 1)

# Print the result in an R Markdown document
cat("Probability of loan acceptance given CC=1 and Online=1\n")
```

```
## Probability of loan acceptance given CC=1 and Online=1
```

```
cat("P(Loan = 1|CC = 1, Online = 1):", prob_loan_given_CC1_Online1, "\n")
```

```
## P(Loan = 1|CC = 1, Online = 1): 0.1071429
```

4. Pivot table for Loan as a function of Online

```
ftable(train.df$Personal.Loan, train.df$Online)
```

```
##       0    1
##
## 0  1102 1620
## 1    99  179
```

5. Compute the following quantities

```
prob_CC1_given_Loan1 <- sum(data$CreditCard == 1 & data$Personal.Loan == 1) / sum(data$Personal.Loan == 1)
prob_Online1_given_Loan1 <- sum(data$Online == 1 & data$Personal.Loan == 1) / sum(data$Personal.Loan == 1)
prob_Loan1 <- sum(data$Personal.Loan == 1) / nrow(data)
prob_CC1_given_Loan0 <- sum(data$CreditCard == 1 & data$Personal.Loan == 0) / sum(data$Personal.Loan == 0)
prob_Online1_given_Loan0 <- sum(data$Online == 1 & data$Personal.Loan == 0) / sum(data$Personal.Loan == 0)
prob_Loan0 <- sum(data$Personal.Loan == 0) / nrow(data)

# Print the results
cat("P(CC = 1|Loan = 1):", prob_CC1_given_Loan1, "\n")
```

```
## P(CC = 1|Loan = 1): 0.2979167
```

```
cat("P(Online = 1|Loan = 1):", prob_Online1_given_Loan1, "\n")
```

```
## P(Online = 1|Loan = 1): 0.60625
```

```
cat("P(Loan = 1):", prob_Loan1, "\n")
```

```
## P(Loan = 1): 0.096
```

```
cat("P(CC = 1|Loan = 0):", prob_CC1_given_Loan0, "\n")
```

```
## P(CC = 1|Loan = 0): 0.2935841
```

```
cat("P(Online = 1|Loan = 0):", prob_Online1_given_Loan0, "\n")
```

```
## P(Online = 1|Loan = 0): 0.5957965
```

```
cat("P(Loan = 0):", prob_Loan0, "\n")
```

```
## P(Loan = 0): 0.904
```

6. Naive Bayes probability P(Loan = 1|CC = 1, Online = 1)

```
prob_loan_given_CC1_Online1_naive_bayes <- (prob_CC1_given_Loan1 * prob_Online1_given_Loan1 * prob_Loan1) / (prob
_CC1_given_Loan1 * prob_Online1_given_Loan1 * prob_Loan1 + prob_CC1_given_Loan0 * prob_Online1_given_Loan0 * prob
_Loan0)

cat("6. **Naive Bayes Probability:**\n\n")
```

```
## 6. **Naive Bayes Probability:**
```

```
cat("P(Loan = 1|CC = 1, Online = 1): ", prob_loan_given_CC1_Online1_naive_bayes, "\n")
```

```
## P(Loan = 1|CC = 1, Online = 1):  0.09881706
```

7. Compare this value with the one obtained from the first pivot table. Which is a more accurate estimate?

```
cat("### 7. Comparison of Estimates\n")
```

```
## ### 7. Comparison of Estimates
```

```
# Print the probabilities for comparison
cat("P(Loan = 1|CC = 1, Online = 1) from the first pivot table:", prob_loan_given_CC1_Online1, "\n")
```

```
## P(Loan = 1|CC = 1, Online = 1) from the first pivot table: 0.1071429
```

```
cat("Naive Bayes Probability (P(Loan = 1|CC = 1, Online = 1)):", prob_loan_given_CC1_Online1_naive_bayes, "\n")
```

```
## Naive Bayes Probability (P(Loan = 1|CC = 1, Online = 1)): 0.09881706
```

```
# Compare the two estimates
if (prob_loan_given_CC1_Online1 > prob_loan_given_CC1_Online1_naive_bayes) {
  cat("The estimate from the first pivot table is higher, suggesting it may be more optimistic.\n")
} else if (prob_loan_given_CC1_Online1 < prob_loan_given_CC1_Online1_naive_bayes) {
  cat("The Naive Bayes estimate is higher, indicating it may provide a more conservative prediction.\n")
} else {
  cat("Both estimates are equal.\n")
}
```

```
## The estimate from the first pivot table is higher, suggesting it may be more optimistic.
```

The estimate from the first pivot table is higher, it suggests that the model used to generate that estimate might be more optimistic about the probability of a customer accepting the loan given they have a bank credit card and are actively using online banking services.The value obtained from the crossed pivot table is the more accurate estimate, since it does not make the simplifying assumption that the probabilities are independent. It is feasible in this case because there are few variables and few categories to consider, and thus there are ample data for all possible combinations

8. Fit a naive Bayes model and compute P(Loan = 1|Online = 1, CC = 1)

```
# Fit a Naive Bayes model
naive_bayes_model <- naiveBayes(Personal.Loan ~ CreditCard + Online, data = train.df)

# Create a new data frame for the specific scenario (Online = 1, CC = 1)
new_data <- data.frame(CreditCard = 1, Online = 1)

# Predict the probabilities for Loan = 1
probabilities <- predict(naive_bayes_model, new_data, type = "raw")

# Print the conditional probability
cat("8. Naive Bayes P(Loan = 1|Online = 1, CC = 1):", probabilities[, "1"], "\n")
```

```
## 8. Naive Bayes P(Loan = 1|Online = 1, CC = 1): 0.1156935
```